LESSON 2.3

Describe the Data Using the Median

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the median of a data set and use it to describe what is typical.

WHY THIS MATTERS You're the stats analyst for the school basketball team. Coach needs a report on the team's scoring: in the last 7 games they scored 42, 38, 55, 41, 38, 44, and 40 points. Which average best represents the team's typical performance?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Median *Spanish: Mediana*	The middle number when you put them in order.	2, 6, 9 . . . median is 6
Mode *Spanish: Moda*	The number that shows up most often.	Data: 2, 3, 3, 5, 7 → 3 appears twice (most often), so mode = 3
Outlier *Spanish: Valor atípico*	A number that is much bigger or smaller than the rest.	2, 3, 4, 40 → 40
Data distribution *Spanish: Distribución de datos*	How the data looks: where it sits and how spread out it is.	If most values cluster in the center with fewer at the ends, the distribution is symmetric
Variability *Spanish: Variabilidad*	How spread out the numbers are.	8, 9, 10, 11 has low variability (close together); 2, 10, 25, 40 has high variability (far apart)
Measure of center *Spanish: Medida de tendencia central*	One number that describes a typical value in a data set.	The median is one of them

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Find the median of 7, 2, 9, 4, 6.

1. Order: 2, 4, 6, 7, 9.
2. Cross off 2 and 9, then 4 and 7.
3. One value is left: 6.

Answer: Median = 6

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

New data set, out of order: 13, 3, 7, 10, 7. Put it in order first.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- In order, the values are _____ .
- There are _____ values, so the middle one is in position _____ .
- The median is _____ , which means _____ .
- About half the values are below _____ .

Word bank median | order | least | greatest | middle | half | typical | center | position | count | outlier | data

WATCH OUT!

A common mistake

A common mistake in Mean, Median, and Mode is finding the median without ordering the data first

LESSON 2.3

Describe the Data Using the Median

Level 0 · Most support · every hint printed

MASTERY CHECK

- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

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WHY THIS MATTERS You're the stats analyst for the school basketball team. Coach needs a report on the team's scoring: in the last 7 games they scored 42, 38, 55, 41, 38, 44, and 40 points. Which average best represents the team's typical performance?

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Find the median of 7, 2, 9, 4, 6.

1. Order: 2, 4, 6, 7, 9.
2. Cross off 2 and 9, then 4 and 7.
3. One value is left: 6.

Answer: Median = 6

1 **Circle the letter of the best answer. Show how you know.**

Find the mean of: 10, 14, 8, 12, 16

(A) 10
(B) 12
(C) 14
(D) 60

STEPS TO TRY

1. Mean means the average of all five numbers — you'll need both a total and how many numbers there are.
2. Use the formula: mean = sum of the values ÷ how many values. Count first: there are 5 numbers here.
3. Step 1: add 10 + 14 + 8 + 12 + 16 to get the total. Then divide that total by 5 — the result is the mean.

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

2 **Circle the letter of the best answer. Show how you know.**

Find the median of: 3, 7, 2, 9, 5

(A) 3
(B) 5
(C) 5.2
(D) 7

STEPS TO TRY

1. Median = the middle number, but only AFTER the list is in order. Don't pick the middle of the unsorted list.
2. Rewrite 3, 7, 2, 9, 5 from least to greatest, then count in from both ends until you reach the one value in the exact middle.
3. Start ordering: the smallest is 2, then 3... finish the ordered list of all five values, then circle the 3rd one — that's the median.

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

3 **Circle the letter of the best answer. Show how you know.**

Find the mode of: 4, 7, 4, 9, 2, 4, 8

(A) 2
(B) 4
(C) 7
(D) 9

STEPS TO TRY

1. Mode is about repeats — which value shows up in the list more times than any other?
2. Make a quick tally: write each different number and count how many times it appears in 4, 7, 4, 9, 2, 4, 8.
3. Start your tally: 7 appears once, 9 appears once... keep counting every number and pick the one with the highest tally.

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

4 **Circle the letter of the best answer. Show how you know.**

Find the mean of: 6, 8, 10, 12, 14

(A) 8
(B) 10
(C) 12
(D) 50

STEPS TO TRY

1. Watch out: one tempting answer is just the total of the list. The mean asks you to go one step further.
2. Mean = total ÷ count. There are 5 values in this list, so after adding you must divide by 5.
3. Step 1: add 6 + 8 + 10 + 12 + 14. Step 2: take that total and divide it by 5. The quotient is your mean.

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

EXPLAIN YOUR THINKING

The mean (43) is higher than the median (41). Why? Which better represents a typical game?

SENTENCE STARTER The mean is higher than the median because the high score of ___ pulls the mean ___. The ___ better represents a typical game because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help